

Behavioral Biometric Identification Using MIDI Keyboard Performance Signatures

Blain Polishak
t00723036@mytru.ca

Aidan Evans
t00708916@mytru.ca

Abstract— *Using behavioural biometric authorization system on the Musical Instrument Digital Interface (MIDI) keyboard performance data, the user can be identified. Our approach of identifying a player on digital MIDI keyboard, is by using keystroke recording for data acquisition; and then machine learning, for the final analysis. The system designed is quite capable of successfully identifying individual musicians based on their distinctive rhythmic, motor, and stylistic patterns. Our current work shows xx% accuracy in identifying users. The hidden potential of accessible biometric systems for music information retrieval and behavioral biometric analysis is defined and can be pursued in the future.*

Keywords— *Behavioral Biometrics, MIDI, Performer Identification, Machine Learning, Feature Engineering, Python*

INTRODUCTION

Musical performances, like most forms of artistic expression, are a direct extension of the individual artist, which encapsulate each individual’s unique motor and cognitive behavior. Subtle variations in timing, dynamics, and pacing emerge from the habitual playing style of a performer and can remain consistent across performances. Behavioural biometrics refers to the identification of individuals based on patterns in human activity, and recent advancements in machine learning have enabled these patterns to be quantified and analyzed with increasing accuracy [1], [2], [3]. In the context of piano performances, MIDI data can provide a structured and precise representation of these behaviors, capturing detailed information such as key velocity, note timing, and note duration. Previously cited approaches to performer identification often rely on audio analysis or computer vision techniques, which can be computationally intensive, challenging to implement, and inaccessible to non-experts. By contrast, MIDI-based analysis offers a low-cost and standardized alternative. Furthermore, previously cited studies also instruct their musicians to play the exact same song, in the same key, at the same BPM, leading to more consistent data, but greatly reducing their system’s adaptability and applicability to realistic use-cases. This project proposes the development of a machine learning system that leverages MIDI performance data to identify musicians based on their distinctive behavioural signatures, trained exclusively by analyzing a musicians freestyle play over a common jazz standard (Autumn Leaves) to a target BPM (90).

LITERATURE REVIEW

Many researchers have investigated music feature selection.

Stamatatos and Widmer [4] investigates the identification of performers in their paper “Automatic Identi-

tification of Music Performers with Learning Ensembles”. They tried the identification across different musical pieces. The authors used recordings of 22 pianists performing two pieces by Frédéric Chopin to analyze and extract stylistic features of their performances [4]. These features are then used within an ensemble of classifiers to identify the performer based on expressive characteristics such as timing and dynamics. Their results show that performers can be recognized even when playing different pieces, demonstrating that certain stylistic traits remain consistent across performances [4]. This finding is relevant to this current research, because it shows that a robust set of traits can be created to build a model with.

“PiJAMA: Piano Jazz with Automatic MIDI Annotations” is another work by Edwards et al. [5], where they introduced a large-scale dataset designed to support research in jazz piano analysis and music information retrieval. The dataset contains over 200 hours of solo jazz piano recordings automatically transcribed into MIDI, comprising 2,777 performances by 120 different pianists across 244 albums [5]. The authors also include automatic audio tagging to identify elements such as applause or spoken introductions, helping clean the data for computational analysis. They analyze statistical properties of the resulting MIDI data, such as pitch distributions and chromaticism, and demonstrate benchmark tasks including performer identification and generative modeling. This work highlights the value of large, automatically annotated datasets for studying stylistic differences and training machine learning models on performance data. Similarly, this project focuses on identifying performers through musical characteristics, but it specifically examines freeform jazz interpretations of the same piece where improvisational choices may reveal distinctive stylistic traits.

In another paper “Pianist Identification Using Convolutional Neural Networks,” Tang et al. [6] used machine

learning to identify the piano playing habits of six professional performers. The authors train a convolutional neural network on MIDI recordings to capture stylistic differences between pianists by analyzing timing, velocity, and other performance characteristics. Their results demonstrate that individual performers can be identified based on consistent patterns in their playing style. This work supports the idea that musical performance contains identifiable behavioral traits[6], which is relevant to this project. Similarly, the goal of this research is to identify performers, but it focuses specifically on distinguishing players through freeform jazz interpretations of the same piece, where individual improvisational style may provide distinctive behavioral features.

The study “Performer Identification From Symbolic Representation of Music Using Statistical Models” by Syed et al. focuses on identifying individual piano performers by analyzing their expressive playing styles using symbolic music (MIDI) data[7]. The authors collected performances of Schubert’s “Sonata in B-flat major, D.960” played by nine virtuoso pianists, capturing features such as timing deviations, dynamics, and note articulation. They introduced a “norm performance” as a reference and computed deviations for each performer. For classification, they used statistical models including Histograms, Kernel Density Estimation (KDE), and Gaussian Mixture Models (GMMs), both for individual features and fused feature sets [7]. Their results show that feature fusion, particularly combining inter-onset interval, dynamic level, and note duration deviations, achieves high precision (0.903) in performer identification, demonstrating the effectiveness of symbolic feature analysis in capturing individual musical style.

In the work, Rhythmic Qualities of Jazz Improvisation Predict Performer Identity and Style in Source-Separated Audio Recordings, Cheston et al. [8] investigate whether rhythmic features alone can be used to identify jazz performers based on recordings. The authors extract rhythmic measures such as ensemble synchronization and rhythmic complexity from 300 recordings of ten jazz pianists, then train a supervised learning model to classify each performer [8]. Their classifier correctly predicts performer identity about 59 % of the time, far above chance and demonstrating that rhythmic behavior contains distinctive signatures of individual style. They also find that clusters of performers share similar rhythmic traits and that each performer’s rhythmic style remains relatively consistent over time [8]. This work highlights that machine learning applied to expressive rhythmic features can reveal individual stylistic behavior in music performance, supporting the feasibility of behavioral biometric identification.

There is no work found on MIDI data capture directly

like our plan. The current research focus is on, whether similar identification is possible within improvisational interpretations of a single piece, where variations arise from spontaneous stylistic choices instead of changes in composition.

MUSICIAN IDENTIFICATION SYSTEM

To build the musician identification system, we needed to develop our own dataset for the musicians playing the songs on the MIDI board.

Required Materials

1) Physical Materials

- MIDI Keyboard (standardized model for all participants)
- Computer (data recording, preprocessing, and training of machine learning models)
- Musicians (data acquisition)

2) Software Materials

- MIDI Recording Software (FL Studio)
- Participant Dataset (recorded MIDI performances of musicians)
- Machine Learning Software (Anaconda/Jupyter Notebook)

Data Collection and Recording Protocol

MIDI data was recorded and exported in a structured format, containing note events, timestamps, velocity, and pitch information. All recordings were performed using the same hardware and software configuration to ensure consistency.

The participants performed improvisations on a predefined jazz standard (Autumn Leaves) with each session consisting of approximately 10 minutes of continuous performance. These recordings were segmented into fixed-length intervals (10–30 seconds), and served as individual samples for analysis. All data was collected under consistent recording conditions to minimize variability unrelated to performer behavior.

Feature Extraction

Raw MIDI performance data, including note timing, information, velocity, and duration was processed to extract meaningful behavioral features. These features collectively form a structured representation of each performance segment.

Raw Features

The following features were given by the raw MIDI data:

- **event_id**: Unique identifier for each MIDI event
- **time_ms**: Timestamp of the note-on event in milliseconds
- **tick**: MIDI tick timestamp (pulses per quarter note)
- **note**: MIDI note number (0-127)

- **note_name**: Note name (e.g. C4)
- **velocity_on**: Note-on velocity
- **time_off_ms**: Timestamp of the note-off event in milliseconds
- **velocity_off**: Note-off velocity
- **duration_ms**: Note duration in milliseconds

Extracted Features

The following features were extracted from each MIDI performance window:

- **avg_velocity**: Mean note-on velocity (loudness)
- **avg_velocity_off**: Mean note-off velocity (release intensity)
- **avg_ioi**: Average inter-onset interval between consecutive notes (ms)
- **avg_duration**: Mean duration of notes (ms)
- **avg_articulation**: Average ratio of note duration to IOI (legato vs. staccato)
- **pitch_range**: Difference between highest and lowest pitch
- **avg_pitch**: Mean pitch value of all notes
- **avg_pitch_step**: Average absolute pitch change between consecutive notes
- **num_chords**: Number of detected arpeggio-like chord sequences
- **avg_notes_per_chord**: Mean notes per detected chord/arpeggio
- **chord_density**: Number of chords per second
- **avg_polyphony**: Average number of simultaneously sounding notes
- **avg_arpeggiation_speed**: Mean time between successive arpeggio notes (ms)
- **avg_arpeggiation_distance**: Mean pitch interval between successive arpeggio notes
- **num_grace_notes**: Count of detected grace notes
- **avg_grace_duration**: Mean duration of grace notes (ms)
- **avg_grace_interval**: Mean pitch interval between grace notes and following notes

Model Building & Evaluation

Data Preprocessing

Initially, each dataset was individually collected and processed to extract the aforementioned compound features. Next, these individual datasets were aggregated into a stacked dataset, comprising all performance windows from each performer, with individual performers being differentiated by userID. This dataset then had its null values replaced with zeros and was divided using an 80/20 training and testing split.

Model Building

Three distinct models were selected to classify and evaluate the test data in order to compare performance across multiple approaches. The first model employed a soft voting ensemble consisting of Extra Trees, K Nearest Neigh-

bors, and Logistic Regression. Feature scaling was applied within the K Nearest Neighbors and Logistic Regression components using StandardScaler.

The second model utilized a preprocessing and classification pipeline, incorporating feature standardization using StandardScaler, dimensionality reduction via Principal Component Analysis (PCA), retaining 90% of the variance, and classification with Logistic Regression.

The third model employed a stacking approach, combining Random Forest and K Nearest Neighbors as base learners, with a Logistic Regression classifier acting as the meta-learner to produce the final predictions.

Model Evaluation

Model performance was primarily evaluated using classification accuracy and confusion matrices. Initial evaluation accuracy was high, with models 1 and 2 scoring 94.87% accuracy and model 3 scoring 92.31%. However, these values were likely overinflated, as an 80/20 split is a suboptimal method of evaluating continuous-performance data. In this approach, portions of the same performance can appear in both the training and testing sets, resulting in minimal behavioural variation between them, particularly when compared to distinct performances collected during separate sessions.

To obtain a more reliable evaluation, the models were tested on entirely new performances and tasked with predicting the corresponding users. In this stage, each model generated class probabilities for individual behavioural time windows within a performance using the predict_proba function. These probabilities were then aggregated using temporal averaging (mean pooling) to produce a single probability distribution per recording, from which the final predicted user was selected as the class with the highest mean probability. The models evaluated 7 performances in total, including 5 from a primary subject. This primary subject collected these samples days apart, and attempted to vary their style and timing for 3 of these sessions.

All models correctly identified the performer for each of the 7 performance datasets producing an average confidence score of approximately 80%, with lower values of 65% observed for the most challenging dataset, which included artificially perturbed inputs. No decision threshold was applied, and each model selected the class with the highest predicted probability. Consequently, the minimum possible confidence for a selection in this step would correspond with a uniform class distribution, equal to 1 divided by the total number of performers in the dataset.

RESULTS

The results of this project provide strong evidence that freestyle jazz improvisation can function as a viable be-

havioural biometric. Unlike controlled performance settings, where musicians replicate the same piece with minimal variation, this study demonstrates that individual identity can still be preserved and recognized even when performers make spontaneous musical decisions. Despite variations in rhythm, phrasing, and note selection across sessions, each performer exhibited consistent underlying patterns in timing, dynamics, and articulation that the models were able to capture. Notably, even when the primary subject intentionally altered their playing style, the system continued to correctly identify them, indicating that deeper motor and cognitive traits persist beyond surface-level stylistic changes. This suggests that improvisation contains stable, performer-specific signatures. As a result, the project highlights the potential of unconstrained musical expression as a rich and underexplored source of behavioural biometric data.

THEORIZED IMPROVEMENTS & FUTURE WORK

There are a number of improvements that could be made to enhance the accuracy and scope of the project. The

biggest improvement would be the collection of a larger dataset. At only 5 participants, the model can only be made accurate to a certain extent. A larger dataset would allow for more rigorous model testing and selection. Additionally, an Aftersense-enabled MIDI keyboard could be used, which would output continuous pressure readings. This would provide lots of useful and detailed pressure data, but it would also increase the cost of deployment. Another notable improvement would be adding an extra set of features. Currently, feature extraction relies heavily on subconscious motor patterns, and there are many more stylistic features that could be included. Right now, only the presence of grace notes and arpeggiation are detected. Additional features could include pedal timing, percentage of in/out of key notes, and average melodic pitch direction. Finally, the system could also be implemented into a live application that allows for automated enrollment and identification. This would give the project more practical usability.

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